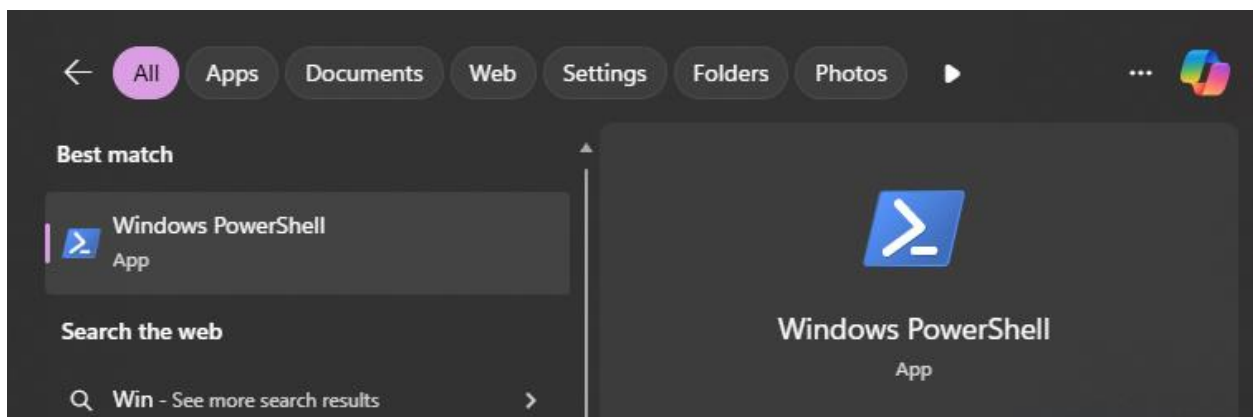


How to Set Up the Miniconda Python Package Manager on a Windows Machine

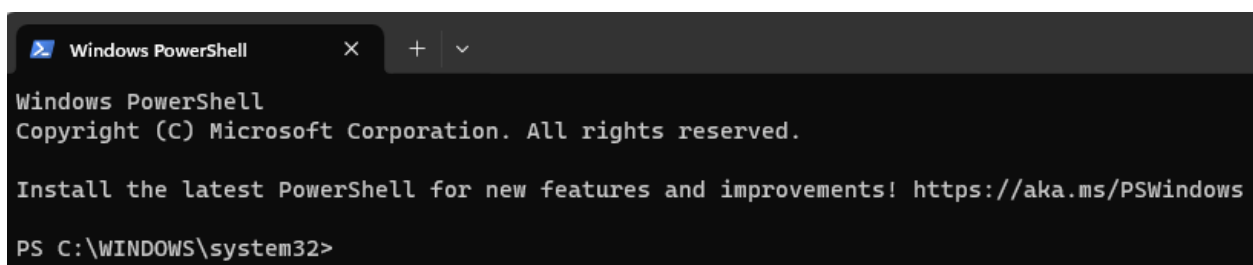
“Miniconda” is a light-weight distribution of the Conda package manager for Python. It’s a very powerful tool for organizing Python environments, easily installing Python-based scientific software packages, and ensuring compatibility across multiple packages being used at the same time.

Step 1: Installing Miniconda

The simplest way to install Miniconda is through the Windows PowerShell terminal. From the desktop interface, type “Windows PowerShell” into the search bar next to the Windows start menu.



This should open a Windows PowerShell terminal looking something like this:



Your command line “path” should read something like “PS C:\Users\username>”; however, if it instead displays “PS C:\WINDOWS\system32>”, enter the following command into the terminal to change your working directory to your home directory:

```
cd ~
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> cd ~
PS C:\Users\computer7>
```

Copy and paste this command into the PowerShell terminal and press enter. This command downloads the latest official Miniconda installer and saves it to your Downloads folder.

```
Invoke-WebRequest -Uri "https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe" -outfile ".\Downloads\Miniconda3-latest-Windows-x86_64.exe"
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Writing web request
Writing request stream... (Number of bytes written: 4796907)

PS C:\Users\computer7> Invoke-WebRequest -Uri "https://repo.anaconda.com/miniconda/Miniconda3-latest-Windows-x86_64.exe" -outfile ".\Downloads\Miniconda3-latest-Windows-x86_64.exe"
```

After the installer finishes downloading, execute the installer silently with all default options via this command:

```
Start-Process -FilePath ".\Downloads\Miniconda3-latest-Windows-x86_64.exe" -ArgumentList "/S" -Wait
```

```
PS C:\Users\computer7> Start-Process -FilePath ".\Downloads\Miniconda3-latest-Windows-x86_64.exe" -ArgumentList "/S" -Wait
Welcome to Miniconda3 py313_25.9.1-3

By continuing this installation you are accepting this license agreement:
C:\Users\computer7\miniconda3\EULA.txt
Please run the installer in GUI mode to read the details.

Miniconda3 will now be installed into this location:
C:\Users\computer7\miniconda3

Unpacking payload...
Setting up the package cache...
Setting up the base environment...
Installing packages for base, creating shortcuts if necessary...
Setting installation directory permissions...
Done!
PS C:\Users\computer7>
```

Since Miniconda is now installed, this next command deletes the installer so that it doesn't take up storage on your computer:

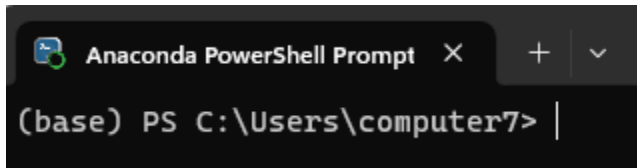
```
del .\Downloads\Miniconda3-latest-Windows-x86_64.exe
```

You can now close this PowerShell terminal.

Step 2: Creating a Conda Environment

2a. Now that Miniconda is installed, we can open the Anaconda Powershell Prompt which should be included with the installation. Type “Anaconda Powershell Prompt” into the search bar and press enter. Your prompt should display something that looks like:

```
(base) PS C:\Users\username>
```

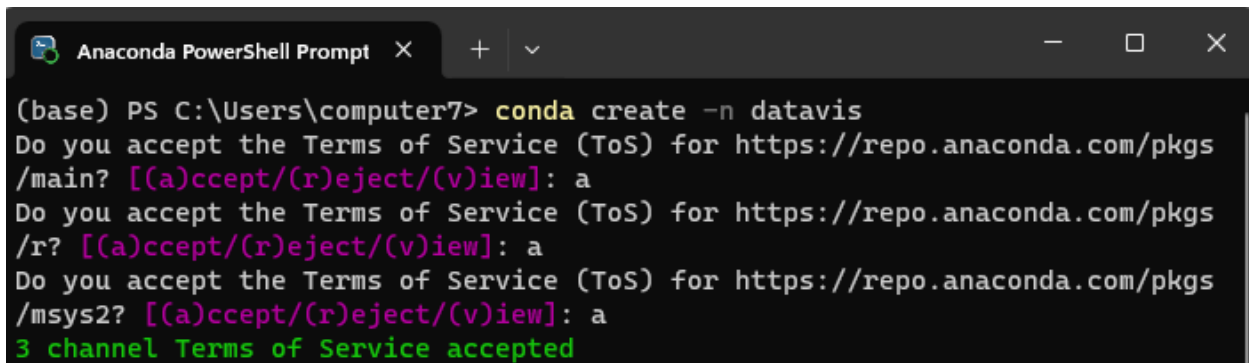


*Note: The Anaconda Powershell Prompt is a modified PowerShell terminal that makes Miniconda automatically available.

2b. The Anaconda Powershell Prompt should have the “base” Conda environment activated by default indicated by the (base) term shown at the beginning of the command line. Now, we’re going to create a custom Conda environment to contain all the packages we’ll be using. Enter the following command into your Anaconda Powershell Prompt to create a new environment called “datavis” for this tutorial:

```
conda create -n datavis
```

Enter “a” when prompted to accept the Anaconda Terms of Service (or read them by entering “v”)



After Conda displays the environment installation path, enter “y” to proceed with the final environment set up.

```
Retrieving notices: done
Channels:
  - defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

   environment location: C:\Users\computer7\miniconda3\envs\datavis

Proceed ([y]/n)? y
```

After a short time, the terminal should display something like the following:

```
Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate datavis
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) PS C:\Users\computer7>
```

We can now activate our new environment using the following command:

```
conda activate datavis
```

You should now notice that the (base) term at the start of your command line now displays (datavis) indicating that we are currently working within our newly created environment.

```
(base) PS C:\Users\computer7> conda activate datavis
(datavis) PS C:\Users\computer7>
```

Additionally, you can view all created environments by entering the following command:

```
conda env list
```

The output of this command should look something like this:

```
(datavis) PS C:\Users\computer7> conda env list

# conda environments:
#
# * -> active
# + -> frozen
base                C:\Users\computer7\miniconda3
datavis              * C:\Users\computer7\miniconda3\envs\datavis
```

Note: the currently activated environment is indicated with a “” before the environment’s installation path

Step 3: Installing Python Packages for Data Visualization

Here, we configure our environment with the Python packages most used for data management and visualization. With the datavis environment activated, we can run the following command to install these packages only into the datavis environment:

```
conda install ipykernel pandas numpy matplotlib seaborn
```

A long list of package dependency names should appear in the terminal followed by this prompt:

```
Proceed ([y]/n)? y
```

Entering “y” will initiate the download and install process like so:

```
Anaconda PowerShell Prompt
Downloading and Extracting Packages:
mkl-2025.0.0      | 104.7 MB | ##### | 100%
icu-73.1         | 29.5 MB  | ##### | 100%
qtdeclarative-6.9.2 | 25.0 MB  | ##### | 100%
qtbase-6.9.2     | 23.5 MB  | ##### | 100%
python-3.14.0    | 15.8 MB  | ##### | 100%
pandas-2.3.3     | 13.6 MB  | ##### | 100%
contourpy-1.3.3  | 11.3 MB  | ##### | 100%
matplotlib-base-3.10 | 8.0 MB   | ##### | 100%
qttools-6.9.2    | 6.5 MB   | ##### | 100%
pywin32-311      | 6.5 MB   | ##### | 100%
numpy-base-2.3.5 | 6.5 MB   | ##### | 100%
aom-3.12.1       | 5.5 MB   | ##### | 100%
pygments-2.19.2  | 4.7 MB   | ##### | 100%
zeromq-4.3.5     | 4.0 MB   | ##### | 100%
libpq-17.6       | 3.8 MB   | ##### | 100%
fonttools-4.60.1 | 3.6 MB   | ##### | 100%
debugpy-1.8.16   | 3.5 MB   | ##### | 100%
pyqt-6.9.1       | 3.2 MB   | ##### | 100%
libhwloc-2.12.1  | 2.3 MB   | ##### | 100%
intel-openmp-2025.0. | 2.1 MB   | ##### | 100%
setuptools-80.9.0 | 1.9 MB   | ##### | 100%
mysql-libs-9.3.0 | 1.4 MB   | ##### | 100%
libglib-2.84.4   | 1.4 MB   | ##### | 100%
ipython-9.7.0    | 1.2 MB   | ##### | 100%

Preparing transaction: done
Verifying transaction: done
Executing transaction: |
```

Conda automatically installs the version of Python that is compatible with each of these packages and ensures that all packages are compatible with each other. After successful installation, the following message should appear in the terminal:

```
Downloading and Extracting Packages:

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
```

Finally, we can verify the installation of our required packages by entering the following command:

```
conda list
```

```
(datavis) PS C:\Users\computer7> conda list
# packages in environment at C:\Users\computer7\miniconda3\envs\datavis:
#
# Name                               Version           Build                Channel
```

This will output a very large list of packages, but we can look specifically for the ones we know we need:

ipykernel	6.31.0	py314h56ec5ee_0
matplotlib	3.10.6	py314haa95532_1
pandas	2.3.3	py314hdbada53_1

numpy	2.3.4	py314h0be4137_1
seaborn	0.13.2	py314haa95532_3

You now have a portable, modular Python environment equipped for data visualization locally installed on your Windows machine!

Some additional notes:

This Conda environment is compatible with just about any modern IDE; we will be using VSCode moving forward, but you can activate this environment inside of other popular IDEs such as PyCharm, JupyterLab, and Spyder, or you can open an interactive session of Python including all the installed packages through the command line by simply calling “python” while the environment is active as such:

```
(datavis) PS C:\Users\computer7> python
Python 3.14.0 | packaged by Anaconda, Inc. | (main, Oct 22 2025, 08:58:42)
[MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import pandas as pd
>>> import matplotlib.pyplot as plt
>>> import numpy as np
>>> import seaborn as sns
>>> print('Hello World!')
Hello World!
>>> |
```

Installing additional packages is as easy as calling “conda install <package-name>” while our environment is active. You can also create separate environments including different packages for different purposes to keep compatibility stable and organize your Python workflows!